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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2023
Course Title: Physical Chemistry III (*Selected Topics in Physical Chemistry*)
Course: Chem. 310, Full Marks: 70, Time: 4 hrs.

[Answer any five questions]

1. (a) What do you mean by polar and nonpolar molecules? 1.5
(b) Explain the terms distortion polarization and orientation polarization. 2.0
(c) Derive an expression for Clausius-Mosotti equation. 4.0
(d) Describe the Ebert's method and vapour-temperature method for the determination of dipole moment. 3.5
(e) The total molar polarizations of CH_3Cl are 90.12 cc/mole at 1/T = 0.004 and 50.0 cc/mole extrapolated to 1/T = 0.0. Calculate the polarizability and dipole moment for CH_3Cl. 3.0

2. (a) Explain the terms: diamagnetic, paramagnetic and ferromagnetic substances 3.0
(b) Derive an expression of Curie-Weiss law for paramagnetic substance. 4.0
(c) Describe the Gouy's method for the determination of magnetic susceptibility. 4.0
(d) The molar magnetic susceptibility of liquid water at 25 ^{0}C is -13×10^{-12} m^3 mol^{-1}, while density is 0.9970 g cm^{-3}. Calculate (i) the specific magnetic susceptibility and (ii) magnetic permeability of the liquid water at this temperature. 3.0

3. a) Explain the differences between thermochemical and photochemical reactions. 2
b) State and explain Grotthus law of photochemistry. 2
c) Explain the fundamental principle for the generation of LASER radiation with example. 4
d) Discuss chemiluminesence with a suitable example. 3
e) A certain system absorbs 3×10^{18} quanta of light per second. On irradiation for 20 minutes 0.003 mole of the reactant was found to have reacted. Calculate the quantum yield for the process (Avogadro's number 6.023×10^{23}). 3

4. (a) What do you mean by catalyst and catalysis? 1.5
(b) Explain the terms autocatalysis, catalytic poisons and catalytic promoters with suitable examples. 3.0
(c) Write down the reactions between CO and H_2 in presence of different catalysts. 1.5
(d) Derive an expression for kinetics of bimolecular heterogeneous surface reactions. 5.0
(e) Derive an expression for specific rate constant using absolute rate theory in heterogeneous gas reaction. 3.0

5. (a) What do you mean by physisorption and chemisorption? 2.0
(b) Explain different types of physical adsorption isotherms. 3.0
(c) What are the assumptions of Langmuir adsorption isotherm? Derive an expression for Langmuir adsorption isotherm. 4.0

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(d) What do you mean by surface excess and surface free energy? 2.5

(e) A liquid of density 997.1 kg m^{-3} rises through 7.36 cm in a capillary of radius 0.2 mm at 25 ^{0}C. Calculate surface tension of that liquid at this temperature. 2.5

6. a) What is colloidal dispersion? 1

b) Define lyophilic and lyophobic sols. 2

c) Discuss Brownian motion of colloids 3

d) Explain the cause of stability of colloidal solution. How could they be precipitated? 4

e) Describe the theory of electrical double layer. What is Zeta potential? 4

a) Explain the terms , i) Phase ii) Component iii) Degree of freedom 4.5

b) Calculate the number of phase, components and degree of freedom in the following system: 2

$$N_2O_4 \rightleftharpoons 2NO_2$$

c) What is Lever rule? 2

d) Define, i) Eutectic point ii) Congruent melting iii) azeotropic mixture 4.5

e) Only draw phase diagram of three component system. 1

8. Write short notes on (any four) of the following: 3.5x4

(a) Photosensitization

(b) Application of colloids

(c) BET isotherm

(d) Incongruent melting point

(e) Emulsions

(f) Resonance fluorescence and quenching of fluorescence

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Department of Chemistry
B.Sc. (Hons.) Part III Examination 2023
Course Title: Physical Chemistry IV (Molecular Motion and Reaction Kinetics)
Course: Chem. 312; Full Mark: 35
Time: 2.5 hours

(Answer any five questions)

1(a)	What is collision frequency? Derive an equation for collision frequency	2
(b)	State Fick's first law of diffusion.	1
(c)	Provide the molecular interpretation for each of the following processes: (i) diffusion, (ii) thermal conduction, (iii) electric conduction, and (iv) viscosity.	4
2(a)	Define ionic mobility and transport number.	2
(b)	Describe the moving boundary method for the measurement of transport numbers.	3
(c)	Discuss the Grotthuss mechanism for the rapid mobility of hydrogen ions in liquid water.	1
(d)	The limiting molar conductivities of KCl, KNO_3, and $AgNO_3$ are 14.99, 14.50, and 13.34 mS $m^2.mol^{-1}$, respectively. What is the limiting molar conductivity of AgCl?	1
3(a)	What is the zero-order reaction? The half-life of a zero-order reaction is proportional to the initial concentration of the reactant-Explain	1+1
(b)	The rate constant of a zero-order reaction is 0.2 mol L^{-1} h^{-1}. After half an hour, the reactant concentration was reduced to 0.05 mol L^{-1}. Calculate the initial concentration of the reactant.	2
(c)	Write the names of the methods for the determination of the order of a reaction. Discuss the half-life method for determination of the order of a reaction	1+2
4(a)	Define consecutive reaction.	1
(b)	What is steady-state approximation? Derive the rate law for the decomposition of N_2O_5 using the steady-state approximation.	4
(c)	Explain unimolecular chemical reactions.	2
5(a)	What is a chain reaction? Derive the rate equation for the formation of hydrogen bromide by a chain reaction.	1+3
(b)	What is an autocatalysis reaction? Discuss the Lotka-Volterra mechanism of an autocatalytic chemical reaction.	1+2
6(a)	Distinguish between diffusion-controlled and activated-controlled reactions.	2
(b)	Describe the formulation of the Eyring equation.	3
(c)	Discuss the chemiluminescence technique used in the study of chemical dynamics.	2
7	Write short notes on any three of the following: (a) Arrhenius equation (b) Thermal explosions (c) Homogeneous catalysis (d) Collision theory (e) Potential energy surfaces	2.5+2.5+2

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Department of Chemistry
Jahangirnagar University
B. Sc. (Hons.) (**Part III**) Final Examination 2023
Course Title: ***Inorganic Chemistry III (Advanced Concepts)***
Course No. Chem. 320F
Time: 4 Hours, Total Marks: 70

Answer any five (5) questions

1. (a) Derive the Schrödinger equation. What is the significance of ψ used in this equation? 3+1
(b) What is meant by (i) Well behaved function (ii) normalization of wave function? 3
(c) Give a brief comparison of particle in one dimensional and three dimensional box calculations. 3
(d) Give the energy expression for a particle of mass 'm' moving in a cubic box of length 'a'. Hence (i) Calculate the zero point energy (ii) Show that the 2nd energy level is degenerate. 4

2. (a) Draw the MO diagram of NO molecule. Compare the bond order and the bond length of NO, NO^+ and NO^-.
(b) Predict the structures of (a) XeF_2, (b) $[XeF_5]^-$ and (c) ClF_3 molecules using the VSEPR model. 3.5
(c) Draw the bonding and antibonding orbital resulted from 2*p* orbitals with (i) σ interaction (ii) π interaction 4.5
(d) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 1.5
(e) Use Slater's rule to estimate values of Z_{eff} for (i) a 4*s* and (b) a 3*d* electron in a V atom ($Z = 23$). 1.5 3

3. (a) Explain coordination number and coordination sphere with suitable example 2
(b) What is the difference between anti-ferromagnetic and ferrimagnetic substances? What is the cause of the difference?
(c) What is effective atomic number (EAN)? Calculate the EAN of the central metal atom of the following complexes and hence comment on their stability. 3
(i) $[Mn(CO)_5]$ (ii) $[Mn(CO)_5]^-$ 1+2=3
(d) Draw the structure of
(i) Linkage isomers of $[Co(NH_3)_5NO_2]^{2+}$ 2
(ii) Ionization isomers of $[Co(en)_2(NCS)(Cl)]NO_2$
(e) Explain the formation of the following on the basis of VBT:
(i) Diamagnetic $[Ni(CN)_4]^{2-}$ (ii) tetrahedral $[NiCl_4]^{2-}$ 4

4. (a) Explain the splitting of *d* orbitals of metals in coordination compounds. Briefly discuss the factors that affect the crystal field splitting energy (Δ).
(b) Which *complex* of the following pairs of complexes has a larger Δ value? Explain your answer: 2+3
(i) $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$, (ii) $[CoCl_4]^{2-}$ and $[Co(NH_3)_4]^{2+}$, (iii) $[Co(H_2O)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$. 4.5
(c) Determine the number of unpaired electrons and calculate the $CFSE_t$ (ignore the pairing energy) of the following: $[Cr(H_2O)_6]Cl_2$, $[Ni(NH_3)_6]^{2+}$, and $K_3[Co(CN)_6]$. 4.5

5. (a) What do you understand by *trans effect* and *trans influence*? Do they have any ***differences*** if have, explain with suitable examples? 3
(b) Explain the reaction mechanism of the substitution reaction in *trans* and *cis*-$[M(LL)_2BX]$ (LL = a bidentate ligand such as en) complexes. 6
(c) Draw a molecular orbital energy level diagram for an octahedral complex $[CoL_6]^{3-}$ showing bonding, non-bonding and anti-bonding MOs. 3
(d) How many geometric isomers of the $[CoBr_2(en)(NH_3)_2]^+$ ion are possible? 2

6. (a) Explain the effect on the *d*-orbital energies when an octahedral complex is compressed along the *z* axis. 3
(b) What is the expected ordering of values of Δ_{oct} for $[Fe(H_2O)_6]^{2+}$, $[Fe(CN)_6]^{3-}$ and $[Fe(CN)_6]^{4-}$? Rationalize your answer. 3
(c) Describe and explain the Jahn-Teller effect in octahedral complexes of Cr^{2+} and Cu^{2+}. 3
(d) Calculate the spin only magnetic moment for a d^8 ion in octahedral, square planar and tetrahedral ligand fields. 3

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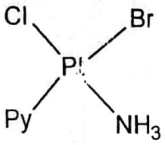
(e) Which complex has more covalency in its metal-ligand bonding, $[Pt(CN)_6]^{2-}$ or $[Cr(en)_3]^{3+}$? Explain. 2

7. (a) What are inner and outer complexes? Inner complexes are stable while the outer complexes are labile- Explain why? 3
(b) Chelate complexes are generally more stable – explain. 3
(c) Octahedral and tetrahedral complexes of same metal have different colour- Explain. 2
(d) Discuss the origin of magnetic property in coordination compounds. 3
(e) Tetrahedral complexes of Cu (II) has larger magnetic moment compared to square planer Cu(II) complexes – Explain why? 3

8. (a) What do you mean by ligand group orbital? 2
(b) Mention the orbitals that are suitable for pi-interaction in octahedral metal complexes. Show by drawing how this interaction affects Δo value. 4
(c) Using K_2PtCl_4 and other appropriate reagents, show by reaction scheme how will you prepare the compound below: (Given that *trans* effect follows the order $Br^- > Cl^- > py > NH_3$ 3



(d) Explain the following: 5
(i) In $[Cu(NH_3)_6]^{2+}$ ion, four Cu-NH_3 bond lengths are 2.07Å and the other two Cu-NH_3 bond lengths are 2.2Å.
(ii) The lability of a complex decreases with the increase of charge on central metal ion.

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) **Part - III** Examination, 2023
Course Title: ***Inorganic Chemistry* (IV) (*Selected Topics*)**
Course No: Chem-322
Full Marks: 35, Time: 2.5 hours

(***Answer Any Five Questions***)

1. (a) Describe the structure and bonding in pure metal and alloys. Explain the properties of pure metals and alloys using your understanding of their structure and bonding. 2+2=4
(b) Describe the band theory of metal with examples. 3

2. (a) Compare the solvent properties of H_2O and liquid NH_3. 2
(b) Discuss the properties of ionizing solvent. 2
(c) Write the self ionization reactions of liquid SO_2 and liquid NH_3 with examples 2
(d) Which of the following non-aqueous solvents has the longest liquid range?
(i) H_2SO_4 (ii) N_2O_4 (iii) NH_3 (iv) HF and why? 1

3. (a) Define electron deficient compound with examples. Compare the structures of C_2H_6 and B_2H_6 compounds. 1+2=3
(b) Write the reactions of B_2H_6 with NH_3 at different conditions 2
(c) Draw the structures of the following compounds:
(i) $Al(BH_4)_3$ and (ii) $Be(BH_4)_2$. 2

4. (a) Explain why the 4f electrons of the lanthanide elements do not take part in chemical bonding. 2
(b) Discuss the methods for the separation of lanthanides from Monazite. 3
(c) Discuss the color of actinide ions. 1
(d) How do the lanthanide complexes act as NMR shift reagent? 1

5. (a) Explain the similarities and differences in structure between myoglobin and hemoglobin in the deoxy and oxy states. 2
(b) Why are the distal and proximal histidines in myoglobin or hemoglobin. 1
(c) What is the oxyhemoglobin dissociation curve? Draw the Standard Oxyhemoglobin Dissociation Curve and explain the factors that affect the standard dissociation curve. 1+2+1= 4

6. (a) What is called polymer? How is polymer classified? 1.5
(b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced? 1.5
(c) Discuss the salient feature of structure and bonding in $[NPCl_2]_3$. 2
(d) Draw the formulae of one polymer having Si–O, B–N and S–S bond. 2

7. Write short notes on the following topics: (any three) 3+2+2=7
(i) Acid-base phenomenon in non-aqueous solvent
(ii) Sulfur-nitrogen polymer
(iii) Chemistry of Fullerenes
(iv) Iron-sulfur proteins
(v) Vitamin B_{12} co-enzymes

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Final Examination 2023
Course Title: Molecular Symmetry and Group Theory
Course No.: Chem-324H
Time: 2.5 hours; Full Marks: 35

[***Answer any FIVE (05) questions***]

1. (a) What is '*principal rotation axis*' of a molecule? Differentiate between '*vertical plane*' and '*horizontal plane*'? When does a molecule possess '*center of symmetry*'? **1 + 1.5 + 1.5 = 4**

(b) Show all the C_2 axes in B_2H_6. **1**

(c) List the four symmetry elements of *cis*-2-butene. **1**

(d) Suppose a molecule belongs to the point group D_{3d}. What information about the molecular symmetry are embedded in the point group symbol D_{3d}? **1**

2. (a) Determine the point group of the following molecules using an appropriate flow chart:

(i) SO_2Cl_2, (ii) $SeCl_4$, (iii) $[C_5H_5]^-$ (iv) *trans*-$C_2H_2Cl_2$ **1.25 × 4 = 5**

(b) Set up the multiplication table for the symmetry operations of pyridine molecule (C_5H_5N). Apply the top operation then the side operation during the process. **2**

3. (a) The term GROUP has a precise mathematical meaning, and the set of symmetry operations of a molecule constitutes a mathematical group. A group consists of a set of members followed some rules. What are those rules? **2**

(b) Taking the C_2 axis as the z axis and σ_h to the xy plane, to what representations do d_{x2-y2}, and d_{z2} orbitals belong in C_{2h} symmetry. The Character Table of C_{2h} point group is given below, in part, for your convenience: **2**

C_{2h}	E	C_2	i	σ_h
A_g	1	1	1	1
B_g	1	−1	1	−1
A_u	1	1	−1	−1
B_u	1	−1	−1	1

(c) Use the following diagram to answer the questions listed below (i and ii):

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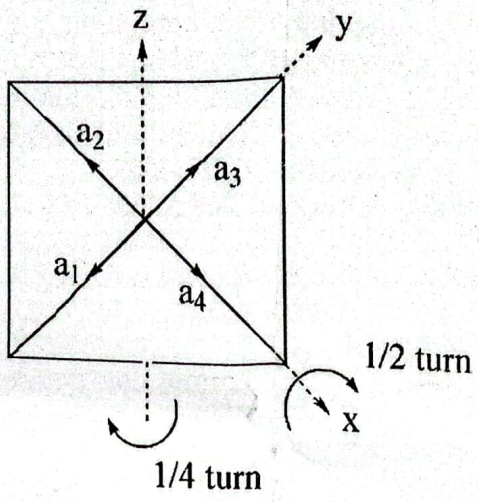

basis: a_1 to a_4

(i) Write down the characters of the matrices representing the quarter turn and half turn rotations, using the four arrows (a_1 to a_4) shown in the above diagram as the basis. **2**

(ii) Write down the character of the matrix representing the operation of reflection in the *yz* plane, using the same basis. **1**

4. (a) What is the order of the dihedral group D_4, which represents the symmetries of a regular square? **1**

(b) Do the following multiplications: **3**

(i)
$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \times \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 5 & 4 & 3 \end{bmatrix}$$

(ii)
$$\begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

(iii)
$$[1 \quad 2 \quad 3] \times \begin{bmatrix} 1 & -1 & -2 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{bmatrix}$$

(c) Identify and draw different types of planes present in CO_3^{2-} ion **2.0**

(d) How do you distinguish between C_{nv} and C_{nh} point group? **1.0**

5. (a) Draw sketches to identify the following symmetry elements **3.0**

(i) a δ_d in C_6H_6 (ii) σ_h in PCl_5 (iii) Both σ and σ' in H_2O (iv) a C_3 axis in NH_3

(v) a C_2 axis in C_2H_4

(b) Draw the structure of the following molecules and assign point group to each using an appropriate flow chart. **4.0**

(i) XeF_4 (ii) SF_4 (iii) BF_3 (iv) PCl_3

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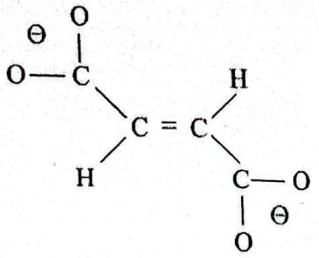
6. Find the number, and symmetry species, of the Raman and infrared active vibrations of the fumarate ion (C_{2h}). The structure of fumarate ion and the Character Table of D_{2h} point group is given below for your convenience: **7**

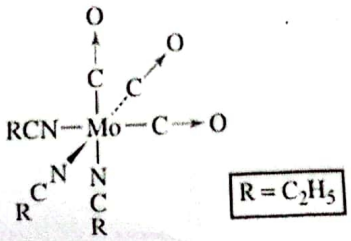


The ion lies in the xy plane. The C_2 axis is the z axis.

C_{2h}	E	C_2	i	σ_h		
A_g	1	1	1	1	R_z	x^2, y^2, z^2, xy
B_g	1	-1	1	-1	R_x, R_y	xz, yz
A_u	1	1	-1	-1	z	
B_u	1	-1	-1	1	x, y	

7. Determine the number of IR-active CO stretching modes for *fac*-$Mo(CO)_3(CH_3CH_2CN)_3$, as shown below. The character table of the C_{3v} point group is given below. **7.0**

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	−1	R_z	
E	2	−1	0	$(x, y), (R_x, R_y)$	$(x^2 - y^2, xy), (xz, yz)$



Department of Chemistry

Jahangirnagar University

B.Sc. (Hons) Part-**III** Examination 2023

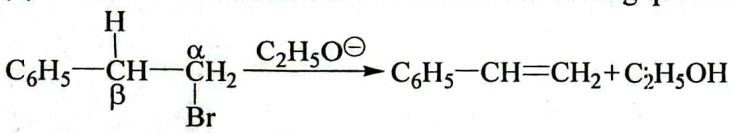
Course Title: ***Organic Chemistry III***

Course No. 330, Time: 4.0 Hours, Total Marks: 70

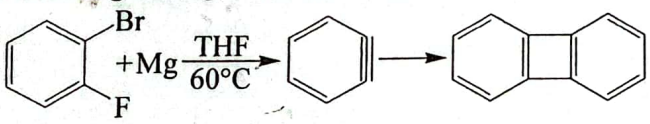
Answer any Five (05) *questions*

1. (a) What is regioselectivity? State a reaction that surely follows regioselectivity. 3
   (b) Why ***trans***-1- phenylpropene give only 70% ***ANTI***-addition product when react with bromine? 4
   (c) The reaction of ***cis***-2-butene with cold alkaline $KMnO_4$ gives 2-methyl-2- butanol. Propose a mechanism showing all intermediates to justify the above statement. 4
   (d) Explain the fact that HBr is the only one of the four hydrogen halides which will add readily to alkenes via a radical pathway. 3

2. (a) Define the terms primary kinetic isotope effects and secondary kinetic isotope effects? 2
   (b) Consider the reaction and answer the following questions:



If the two ***beta*** (β) hydrogen atoms of 1-bromo-2-phenyl ethane are replaced by the deuterium atoms (D) then we have k_H/k_D=7.1.
   (i) Does the reaction follow the primary kinetic isotope effect or the secondary kinetic isotope effect? Give explanation in favor of your answer. 2
   (ii) Does the reaction follow the E2 or E1 mechanism? Explain. 2
   (iii) Write down the mechanism for the reaction considering k_H/k_D = 7.1. 3
   (iv) If k_H/k_D=1.0 in the above reaction then what does it imply? Give explanation in support of your answer. 1
   (c) What is trapping? When does a researcher usually trap an intermediate product to determine the mechanism of a reaction? 1+1=2
   (d) Using the trapping method how can you prove that the following reaction is occurring through the formation of benzyne intermediate? 2



3. (a) Discuss Hofmann and Saytzev elimination rule with suitable example. 4
   (b) Show the mechanism that ***trans***-2-butene gives ***meso***-2,3-dibromobutane as a sole product. 3
   (c) Give two evidences in support of the formation of cyclic bromonium ion during the course of addition of bromine to alkenes. 4
   (d) What is the effect of leaving group on nucleophilic substitution reaction? 3

4. (a) Both the $LiAlH_4$ and $NaBH_4$ are the hydride donors. Which one of the hydride donors will you select to reduce the carbonyl group of an ester function? Give explanation in support of your answer. 2
   (b) Write down the mechanism of the reduction of ester using your selected hydride donor that you have selected in the question no (a). 3
   (c) Give structures of the unknown compounds **A-B** and answer the following questions: 1

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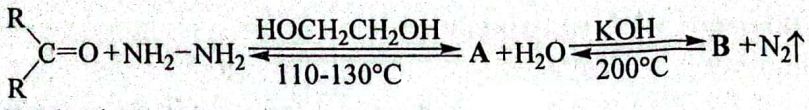


(i)What is the name of the reaction? 0.5

(ii) Why the reaction is carried out in two different temperatures (110-130°C and 200°C)? Give explanation in favor of your answer. 2

(iii) Write down the mechanism of the formation of the product **B** from the product **A** showing the all curved arrows. 3.5

(iv) Write down the limitations of the reaction with suitable examples. 2

5. (a) Why C=O group of ester absorbs at higher frequency than ketone in IR spectrum? 2

(b) Explain that the λ_{max} value of π–π^* transition of 1, 3 butadiene is larger than that of ethylene. 3

(c) Define MacLafertty rearrangement. Show the fragmentation pattern for 2-pentanone. 3

(d) An unknown compound has the following IR and ^{1}H-NMR data, predict the name and structure of this unknown compound and also explain each of the spectral ***value*** 6

IR: ν_{max} 3105, 2920, 2825, 2731, 1690, 1596, 1502, 1390, 1020 cm^{-1}

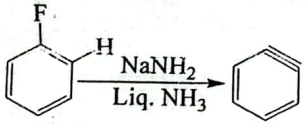
^{1}H-NMR: δppm 3.79 (3H, s), 6.93 (2H, d), 7.81 (2H, d), 9.89 (1H, s)

6. (a) Define E1cB elimination reaction with appropriate example. 1.5

(b) What are basic differences between the E1cB and E1 elimination reactions? 2.5

(c) Write down the factors that are important to occur an E1cB elimination reaction. 2.5

(d) Consider the following reaction and answer the given questions:



(i) What is the name of the product? 0.5

(ii) What are the functions of the fluorine atom in the above reaction? 2

(iii) What is the hybridization of the carbon atoms of the carbon-carbon triple bond of the product molecule. 2

(iv) Briefly discuss about the bonding pattern of the triple bond of the product in the light of orbital picture. 3

7. (a) Discuss the mechanism of the following reactions (any three): 3×3= 9.0

(i) Clemmensen reduction.

(ii) The Baeyer-Villiger oxidation.

(iii) The Meerwin-Ponndorf -Verley reduction.

(iv)) Oppenauer oxidation.

(b) Why the S_N2 reaction is carried out in the presence of approtic polar solvents whereas, the S_N1 reaction is carried out in the presence of protic polar solvent? Give explanation in support of your answer. 2

(c) How does bromine (Br_2) react with alkenes although both the bromine (Br_2) and the reactant alkenes are non-polar molecules? Explain in terms of molecular orbital concept. 3

8. (a) Define relative and absolute configuration. 2

(b) Discus two methods for interconversion of ***cis-trans*** isomers. 4

(c) Which diastereomer of 2, 3-dibromobutane undergoes elimination reaction more rapidly when treated with potassium iodide? Explain your answer. 4

(d) Show the determination of configuration of (+)-glyceraldehyde, (-)-lactic acid and (+)-maleic acid with respect to (+)-tartaric acid. 4

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) ***Part-III*** Examination, 2023
Course Title: ***Chemistry of Natural Products***
Course No.: Chem. 332; Full Marks: 70; Time: 4 hours

[Answer any *Five* questions]

1. (a) What are terpenoids? How are they classified? 1+2
   (b) What product would you expect to be formed when citral is treated with (i) $KHSO_4$, Δ; (ii) Ag_2O and (iii) aq. K_2CO_3? 1×3
   (c) Outline a synthesis of myrcene. 2
   (d) Establish the structure of limonene by degradative and synthetic methods. 4
   (e) Show the biosynthesis of isoprene unit from mevalonic acid. 2

2. (a) Draw the primary structure of DNA and RNA. 4.0
   (b) Find out the structure of uric acid by degradative study. 3.5
   (c) Proof that the sugar molecule is attached in the five-membered ring of the purine base. 3.5
   (d) Why nucleosides are non-reducing? 1.0
   (e) How can you prove that the Medicus structure of uric acid is the actual structure? 2.0

3. (a) Explain why maltose shows mutarotation but sucrose does not. 3.0
   (b) Prove that D-glucose and D-fructose have identical configuration at C-3, C-4 and C-5 positions. 3.0
   (c) How can you synthesize D-glucose from D-arabinose? 2.0
   (d) Describe the evidence for the ring structure of aldose sugar. 3.0
   (e) How can you chemically distinguish between α-D-glucopyranose and β-D-glucofuranose? 3.0

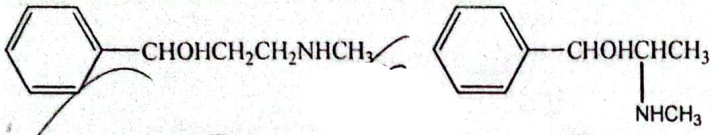
4. (a) Define antibiotics and write down the criteria for being an antibiotic. 3.0
   (b) Elucidate the structure of penicillin. 6.0
   (c) Write down different mechanisms of action, an antibiotic can follow against bacteria. 2.0
   (d) Outline the synthesis of chloramphenicol and write down some unwanted side effects of chloramphenicol. 3

5. (a) Define the terms: essential and non-essential amino acids. Give an example for each class. 2
   (b) How can you synthesize tryptophan from indole using hydantoin method? 3
   (c) Write equations for the reactions of phenylalanine with:
   (i) HNO_2, (ii) NOCl & (iii) $LiAlH_4$.
   (d) Discuss the Edman's method for determining the sequence of α-amino acids in a polypeptide chain. 4
   (e) Give a synthesis of dipeptide, gly-ala using Bergmann method. 2
   (f) What is denaturation of proteins? 1

6. (a) Define alkaloids. What are the main characters of plant alkaloids. 1+1
   (b) How can you detect the following groups in an alkaloid? 3
   (i) -OH (ii) $-OCH_2O-$ (iii) $>N-CH_3$ 1+1+2
   (c) Establish the following facts:

(i) Ephedrine contains a mono-substituted benzene ring.
(ii) Morphine contains a phenanthrene nucleus.
(iii) Tropine contains a pyrrolidine ring.

(d) Which one would be more acceptable structure for ephedrine of the following? Give two evidences in favor of your answer. 3



(I) (II)

(e) Give a synthesis of tropic acid from acetophenone. 2

7. (a) Explain the terms (i) epimer and (ii) mutarotation with suitable examples. 2.0
(b) What is osazone? Show that the D- (+)-glucose and D-mannose give the same osazone. Explain it with proper reactions. 2.0
(c) Distinguish between reducing and non-reducing sugars with suitable examples. 2.0
(d) Write down the reactions when D-glucose reacts with the following reagents: 1×3=3.0
(i) methanol in the presence of HCl; then
(ii) dimethyl sulphate in the presence of sodium hydroxide; then
(iii) dil. HCl.
(e) Describe a method for conversion of an aldose into a higher numbered aldose. 2.0
(f) Discuss about the conformation of (+)-maltose, (+)-sucrose and starch. 3.0

8. (a) What is Diels hydrocarbon? How was the structure of Diels hydrocarbon established? 2.0
(b) What are steroids? What is the basic structural skeleton of cholesterol? Draw the chair conformation of cholesterol and mention the atom numbering. 2.0
(c) Define Blanc's rule and angular methyl group. Explain the following facts during the structure elucidation of cholesterol: 2.0 + (3×2)
(i) Determination of the sizes of A, B and D rings,
(ii) Determination of the position of the hydroxyl group and double bond, and
(iii) Determination of positions of the two angular methyl groups.
(d) What are cardiotonic glycosides and saponins? 2.0

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part III Final Examination 2023
Course Title: Chemical Spectroscopy
Course: Chem. 340 Full Mark: 70 Time: 4 hours

(Answer any five questions)

1. (a) What is electromagnetic radiation? Mention its important characteristics. 3
   (b) Arrange different types of electromagnetic radiation with increasing order of energy. 3
   (c) Why does an atom create a line spectrum whereas a molecule produces a band spectrum? 2
   (d) Discuss important factors that contribute to intensity of spectral lines. 3
   (e) Calculate the energy associated with 1.00 mol of photons of green light (555 nm) [$c = 3.00\times10^{10}$ cm s^{-1}; $h = 6.626\times10^{-34}$ Js]. 3

2. (a) Which of the following molecules will show a microwave rotational spectrum? H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6? 2
   (b) Classify the following molecules on the basis of the principle moment of inertia: HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH. 2
   (c) Write down the energy expression in cm^{-1} for a diatomic molecule based on the rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines. 3
   (d) What are the limitations of the rigid rotator model? Write down the energy expression for the non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra. 4
   (e) Calculate the moment of inertia of the water molecule around the axis defined by the bisector of the HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H=1.0079, atomic mass unit = 1.6606×10^{-27} kg). 3

3. (a) Compare the energy of an anharmonic oscillator and a harmonic oscillator. Calculate the absorption energy of the fundamental band, first overtone, second overtone, and hot band for an anharmonic oscillator. 3
   (b) According to Born—Oppenheimer approximation, derive an expression for a diatomic Vibrating-rotator that undergoes υ=0 to υ=1 transition. 4
   (c) How many fundamental modes of vibration are possible for H_2O molecules? Are all the vibration modes IR active? Explain with reason. 4
   (d) How do you differentiate between CO_2 and N_2O molecules by vibrational spectroscopy, although both are linear triatomic molecules? 3

4. (a) Explain the classical theory of Raman effect with example. 3
   (b) Why all the vibration modes of CO_2 molecule is not Raman inactive althougn it change the polarizability ellipsoid. Justify your statement. 4

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(c) How do you obtain pure rotational Raman spectra? When we get S branch and show stoke and anti-Stokes' lines for S branch. 4

(d) What is meant by the law of mutual exclusion? How do you distinguish between H_2O and HCN using Raman and infrared spectroscopy? 3

5(a) Show the electronic energy level and possible transition for an electron without and with *J*-splitting. What are the reason for *J*-splitting? 4

(b) Write the term symbol for the electrons with a total spin angular momentum of 2, 1, 3/2 and orbital angular momentum of 0, 1, 2 respectively. 3

(c) Which transition of atomic orbital give rises to the compound doublet and triplet spectra? 3

(d) Explain the energy level and the transition of the He atom whose electrons are in a triplet state. 4

6(a) State Franck—Condon principle. Explain the different electronic transitions for a diatomic molecule according to the Franck—Condon principle. 4

(b) What type of electronic transition occurs in the molecule having the following bonds; (i) C—C, (ii) O—H, (iii) C=N. Mention the spectral region of these transitions. 3

(c) Which of the following molecules give rise to the electronic spectra in higher wavelength? Explain why? (i) CH_4 (ii) CH_3NH_2 (iii) $H_2C=CH-CH=CH_2$ 3

(d) Write the name and explain graphically the different ways by which an electronically excited molecule can lose energy. 4

7(a) Deduce an expression to determine the extent of interaction between spin and magnetic field. What is Larmor frequency? 2.5+1

(b) What is meant by chemical shift? How could you distinguish between *cis*- and *trans*-isomer from ^{1}H-NMR spectra? 1+1

(c) Discuss the anisotropic effect on the chemical shift values. 3

(d) Explain why TMS is used as a reference substance in NMR. 1.5

(e) What is ESR? Explain hyperfine structure with an example. 4

8 Write short notes on **any four** of the following: 3.5×4

(a) Signal-to-noise ratio

(b) The stark effect and its application

(c) Principle of Mössbauer spectroscopy

(d) Quantum theory of the Raman effect

(e) ESR spectroscopy

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Department of Chemistry
Jahangirnagar University
B.Sc. (Honors) 3rd year Final Exam 2023
Course: **Chem-350**
Course Title: ***Fundamentals of Pharmaceutical Chemistry***
Full Marks: 35, Time: **2.5 hours**

Answer any Five (5) questions

1. (a) Define API and excipient. Write the examples of the following classes of API. 3
(i) Antiviral agent (ii) Antineoplastic (iii) Antibiotic (iv) Antifungal agents
(b) What is organoleptic properties? Briefly discuss about the different organoleptic properties of pharmaceutical raw materials. 2
(c) What is meant by drug-excipient compatibility? How is drug-excipient compatibility determined? 2

2. (a) Define Prodrug. What is the purpose of using prodrug? 2
(b) Classify receptor according to their mechanism of action. 2
(c) What are the routes of drug administration? 2
(d) What is pharmacokinetics? 1

3. (a) Tablet is the most prescribed doges form. Do you agree? Write the reasons in favor of your answer. 2.5
(b) What is the difference between wet and dry granulation? When dry granulation process is used? 2.5
(c) What is testing? What are the parameters usually tested for tablets? 2.0

4. (a) What is ophthalmic preparation? State the advantage and disadvantages of solution and ointment. 2.5
(b) What is meant by real time and accelerated stability study? Discuss the conditions used for them. 2.5
(c) How is the superior quality of water described? What are tests performed to ensure the quality? 2.0

5. (a) Discuss about mechanism of action sulfonamide drug. 2.5
(b) Write the structure activity relationship of penicillin. 2.5
(c) Outline the synthesis of the following drugs: 2
(i) Sulfacetamide (ii) Sulfamethoxazole

6. (a) What is antifungal agent? How can you synthesis fluconazole? 2.5
(b) What do you mean by HPLC? Write down the application of HPLC in pharmaceutical industry. 2.5
(c) An assay experiment, 35.40 mL of N/10 I_2 solution required to titrate 350 mg of ascorbic acid. The normality factor of iodine solution is 1.108. Calculate the percentage content of L-ascorbic acid in supplied sample. 2

7. (a) Define QA and QC. 2
(b) What are GMP and cGMP? Write down important principles of GMP. 2+1=3
(c) What is the principle of the Karl Fischer titration? How can you calculate the % of water by using Karl Fischer reagent 2

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